Supplement S1

This file contains additional information and analysis for the paper "Conspiracy theories are central to the complex system of predictors of Italian vaccine hesitancy".

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1 Theoretical predictors of vaccine hesitancy worldwide

In this Section, we briefly review the international literature on the determinants of vaccine hesitancy. An important milestone in the modern history of vaccine hesitancy is the notorious article by Wakefield and colleagues, published in 1998. The article triggered a worldwide controversy on the measles–mumps–rubella vaccine, describing it as a potential cause of autism. This study was recently retracted by Lancet, but its effects are still ongoing in the public debate (Poland Gregory & Jacobson Robert, 2011). The COVID-19 pandemic exacerbated the salience of this topic, and between 2019 and 2023, many studies attempted to isolate the determinants of this behavior. Macro-level studies on COVID-19 vaccine hesitancy highlighted that this phenomenon is more pronounced in wealthy countries (Aw et al., 2021; Sallam, 2021). At the micro-level, scholars focused on the individual predictors of vaccine hesitancy. Before the outbreak of COVID-19, an influential work isolated complacency, convenience, and confidence as relevant factors (MacDonald, 2015). Complacency entails a low perception of the risks associated with potential diseases and can lead to considering vaccines as unnecessary. Confidence in vaccines is determined by the degree of trust given to the institutions developing and administering them, and by their perceived effectiveness. Finally, convenience refers to the availability and affordability of these products. More recently, COVID-19-related vaccine hesitancy was analyzed through the epidemiological triad of environmental, agent, and host factors (Sallam, 2021). The first category includes socio-political factors such as national policies, social norms, and media stimuli. The second one comprehends considerations related to vaccine safety and effectiveness. Host factors are individual characteristics, like educational level and income. This classification effectively describes the determinants of hesitancy. However, it is informed by a restricted number of studies, since it was carried out in the preliminary phases of the pandemic.

Hence, this article builds on the theoretical classification proposed by the scoping review of Aw and colleagues (Aw et al., 2021), which scrutinizes 97 peer-review articles published in Medline®, Embase®, CINAHL®, and Scopus® databases between December 2019 and March 2021. The review only includes papers having vaccine hesitancy as the dependent variable, English as the publication language, and high-income countries as the research setting (GNI per capita greater than US $12,535). Their sample mostly features analyses carried out in Europe or North America, with U.S.A. and Italy being the most studied countries. More than 77% of the contributions are based on a cross-sectional design, 14 adopt a mixed-method approach, and only two are based on Randomized Controlled Trials. The authors classify the determinants into Contextual, Individual, and Vaccine related factors. The interested reader is suggested to consult Aw et al. (2021) for an exhaustive list.

Contextual determinants include media factors, political preferences, religion, socio-demographic variables, geographical barriers, and institutional trust. Far-right political attitudes (Head et al., 2020; Reiter et al., 2020; Ruiz & Bell, 2021; Schmelz, 2021; Shekhar et al., 2021) and populist inclinations (Kennedy, 2019; Recio-Román et al., 2021) are associated with hesitancy. Distrust of national governments (Alabdulla et al., 2021; Bogart et al., 2021; Unroe et al., 2021) and of socio-political institutions, (Jennings et al., 2021; Recio-Román et al., 2021) also predict greater hesitancy. Other contextual determinants are individual religiosity in general (Callaghan et al., 2021; Vezzoni et al., 2022), believing in divine immanence in particular (Ladini & Vezzoni, 2022), and the use of digital media as the main source of information (Attwell et al., 2021; Barrière et al., 2021; Feleszko et al., 2021; Group et al., 2020; Jennings et al., 2021). There is an overwhelming consensus on the greater hesitancy of women (37 supporting studies, 12 without statistical significance, only one disconfirmation), and on the negative relationship between educational level and hesitancy (19 supporting studies, 13 without significance) (Aw et al., 2021). The role of income, age, and area of residence is more ambiguous instead. Hesitancy correlates with feminine sex, younger age, and rurality, but the evidence is less conclusive (ibid.). A second class of determinants is associated with group or individual characteristics. Low worry about COVID-19 infection (Bogart et al., 2021; Head et al., 2020; Yoda & Katsuyama, 2021) low perception of risk (Group et al., 2020; Latkin et al., 2021; Malik et al., 2020) and endorsing conspiracy theories (Barrière et al., 2021; Detoc et al., 2020; Schmelz, 2021) all lead to increased hesitancy. Trust in the alternative medicines (Alabdulla et al., 2021; Attwell et al., 2021; Seale et al., 2021), distrust of the healthcare system (Chu & Liu, 2021; Pogue et al., 2020; Sherman et al., 2021) and of science (Gagneux-Brunon et al., 2021; Sherman et al., 2021; Wong et al., 2021) are also associated with hesitancy. Moreover, individual characteristics play an important role. Health locus of control is the degree to which one conceptualizes her/his state of health as either a consequence of personal actions (internal locus of control) or as a product of collective activities (external) (Luszczynska & Schwarzer, 2005). Internal health locus of control (Kouvari et al., 2022; Olagoke et al., 2021), low compliance with preventive measures (Pogue et al., 2020; Schwarzinger et al., 2021) and a lesser sense of collective responsibility are associated with vaccine hesitancy (Barello et al., 2020; Bokemper et al., 2021; Loomba et al., 2021). The last group of determinants is related to vaccines' characteristics. Scholars highlight the role of negative attitudes toward vaccines, such as believing they are unsafe (Motta, 2021; Seale et al., 2021) or too rapidly developed (Alabdulla et al., 2021; Pogue et al., 2020). Fear of needles (Cherian et al., 2022) and disapproval of mandatory vaccination policies (Meier et al., 2019; Rieger et al., 2022) are important factors as well.

2 Node construction

Table 1 presents the Eigenvalues for three Principal Component Analysis models and Cronbach's α of the resulting indexes. Following the Kaiser-Guttman rule, all data structures are unidimensional. We compute mean indexes for variables regarding institutional trust (trust in the Italian and European Parliament), judgments on the Italian government (approval of governmental policies to fight COVID-19, general approval of the Draghi government), and variables tapping compliance to preventive behaviors (wearing face masks, observing social distancing, and washing hands. The resulting indexes are highly reliable. The original survey questions and the descriptives of these variables are reported in Table 2 and Table 3 of the main article.

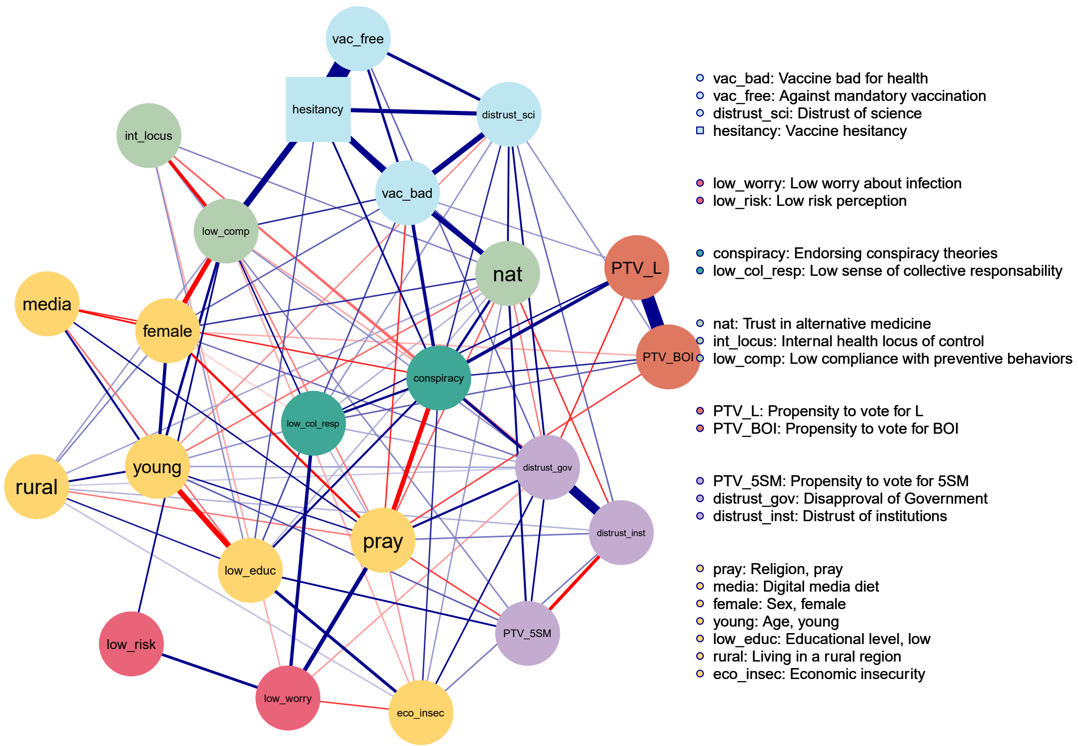
Table 1: Results of PCA

Index	*1°component (λ)*	*2° component (λ)*	*Cronbach's α*
Compliance with preventive behaviors	1.5	0.6	0.8
Disapproval of Government	1.4	0.4	0.9
Distrust of institutions	1.3	0.6	0.8

3 Network estimation

We provide the result of network estimation without the graphical filter omitting edges below 0.06. We chose this value to include in the plot all edges involving the variable hesitancy. Note that the filter is merely a graphical tool. The network plot presented in the main article (Figure 2) presents less noise, but all network metrics were computed on the unfiltered graph presented here. Showing a filtered graph is common in social network analysis, and was done also in other papers adopting a complex system approach to study COVID-19-related issues (Chambon et al., 2021, 2022, 2023). Consistently, Figure 1 below avoids reporting results regarding nodes' percentage of variance explained (presented in the main article through the colored nodes' border). These data are indeed unchanged from the one shown in the main article.

Figure 1: Unfiltered network plot



4 Centrality metrics

Table 2 reports the exact values of Strength and Degree centrality for each node in the network. Strength is calculated by summing the absolute values of all edge weights a node is connected with (Opsahl et al., 2010). Degree counts the number of ties of each node.

Table 2: Strength and Degree Centrality

Variable	*Strength*	*Degree*
Vaccine bad for health	1.61	10
Against mandatory vaccination	1.34	4
Low worry about infection	0.75	6
Low risk perception	0.28	2
Endorsing conspiracy theories	1.83	15
Trust in alternative medicine	1.27	15
Internal health locus of control	0.54	8
Low sense of collective responsability	0.79	11
Propensity to vote for L	1.12	5
Propensity to vote for 5SM	0.74	7
Propensity to vote for BOI	1.01	6
Distrust of science	1.22	10
Religion, pray	1.49	15
Digital media diet	0.48	5
Sex, female	1.01	10
Age, young	1.24	11
Educational level, low	1.09	10
Living in a rural region	0.47	9
Economic insecurity	0.64	9
Vaccine hesitancy	2.13	7
Low compliance with preventive behaviors	1.47	12
Disapproval of Government	1.50	16
Distrust of institutions	1.20	9

5 Community detection

To enhance the stability of the results, we performed a thousand iterations of the Walktrap Community detection algorithm (Pons & Latapy, 2005). Figure 2 and Figure 3 report respectively the result of the iterations and the final output of community detections. Community detection is performed on the absolute adjacency edge weight matrix, thus ignoring the distinction between negative and positive edges.

Figure 2: Iterations of the Walktrap Community detection algorithm

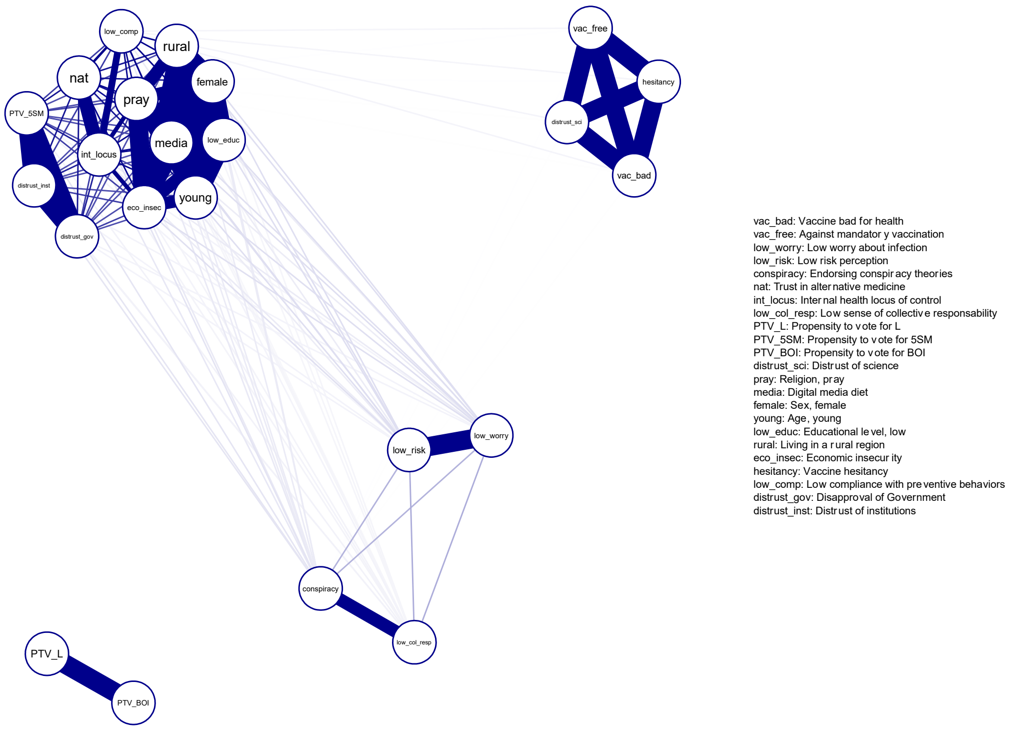
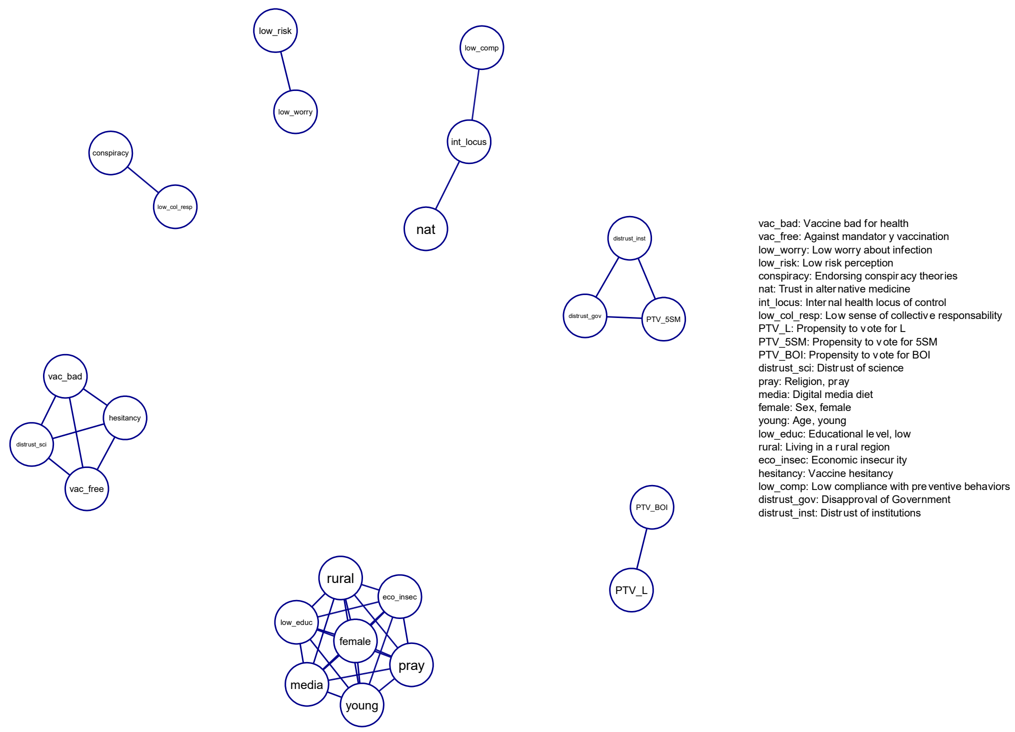


Figure 3: Final results of the community detection algorithm



6 Network weighted adjacency matrix

Figure 4 on the next page reports the full weighted adjacency matrix of the network. Note that 0s (blank cells) are omitted to improve clarity and that only the upper triangle is reported. As network edges are undirected, the bottom triangle is equivalent to the one we show. An Excel file containing the same information is also provided in the GitHub Folder (File: "Output/Supplement/EdgeWeightsExcel_half.xlsx").

Figure 4: Network weighted adjacency matrix

	vac_bad	vac_free	low_worry	low_risk	conspiracy	nat	int_locus	low_col_resp	PTV_L	PTV_5SM	PTV_BOI	distrust_sci	pray	media	female	young	low_educ	rural	eco_insec	hesitancy	low_comp	distrust_gov	distrust_inst
vac_bad		0,15			0,20	0,28			0,04			0,30	-0,08		0,06		0,07			0,34	0,09		
vac_free												0,18								0,95		0,06	
low_worry				0,17			-0,03	0,20					0,23						-0,08			-0,04	
low_risk																					0,11		
conspiracy						0,15	-0,06	0,14	0,19		0,09	0,09	-0,26	-0,09			0,13		0,08	0,12	-0,04	0,04	0,17
nat							0,05	0,03		0,13		0,12	-0,06		0,08	-0,06	0,03	0,04	0,04	0,07		-0,05	-0,08
int_locus										0,05							0,04				-0,20	-0,06	-0,05
low_col_resp									0,10		0,06	0,04	0,03		-0,02	0,04					0,10	0,03	
PTV_L											0,71											-0,08	
PTV_5SM													-0,08			0,07	0,12					0,11	-0,20
PTV_BOI												0,04	-0,08		-0,03								
distrust_sci																-0,04				0,23		0,12	0,06
pray														0,10	-0,14	0,09		-0,05	-0,04		-0,05	0,14	0,05
media															-0,11	0,13	-0,06						
female																0,19		0,05			-0,27	0,06	
young																	-0,30	0,12			0,16	0,04	
low_educ																		0,09	0,17	0,07			
rural																			0,03		0,04	0,02	0,03
eco_insec																					-0,03	0,11	0,05
hesitancy																					0,34		
low_comp																						0,04	
distrust_gov																							0,50
distrust_inst																							

7 Logistic regression models

Here we report on the logistic regression models we performed to evaluate the relationship between Strength scores and regression coefficients. We provide a summary of 17 models that generated the 17 entries of the scatterplot, plus the regression model for the variable "educ", which is used for the final table of empirical predictors (Table 7, main article).

Table 3: Models for distrust_gov, distrust_inst, h_locus, he_eco, low_comp, media

Logit regression model on vaccine hesitancy

	distrust_gov
	Odds Ratios
(Intercept)	0.02 *** (0.01 – 0.04)
distrust_gov	1.43 *** (1.34 – 1.52)
sex	1.39 ** (1.04 – 1.84)
age	1.00 (0.99 – 1.01)
educ	1.78 *** (1.30 – 2.43)
reg	0.98 (0.73 – 1.31)
eco_insec	0.98 (0.78 – 1.22)
Observations	1540
R^2 Tjur	0.125
AIC	1254.662

* *p<0.1* ** *p<0.05* *** *p<0.01*

Logit regression model on vaccine hesitancy

	distrust_inst
	Odds Ratios
(Intercept)	0.01 *** (0.01 – 0.03)
distrust_inst	1.38 *** (1.30 – 1.47)
sex	1.46 *** (1.10 – 1.93)
age	1.01 (1.00 – 1.02)
educ	1.66 *** (1.22 – 2.26)
reg	1.00 (0.75 – 1.32)
eco_insec	1.02 (0.82 – 1.27)
Observations	1540
R^2 Tjur	0.112
AIC	1271.741

* *p<0.1* ** *p<0.05* *** *p<0.01*

Logit regression model on vaccine hesitancy

	h_locus
	Odds Ratios
(Intercept)	0.29 *** (0.13 – 0.65)
h_locus	0.78 *** (0.74 – 0.82)
sex	1.47 *** (1.11 – 1.95)
age	1.00 (0.99 – 1.01)
educ	1.92 *** (1.41 – 2.62)
reg	1.07 (0.81 – 1.42)
eco_insec	1.17 (0.94 – 1.45)
Observations	1540
R^2 Tjur	0.095
AIC	1295.937

* *p<0.1* ** *p<0.05* *** *p<0.01*

Logit regression model on vaccine hesitancy

	he_eco
	Odds Ratios
(Intercept)	0.02 *** (0.01 – 0.05)
he_eco	1.24 *** (1.17 – 1.31)
sex	1.49 *** (1.13 – 1.96)
age	1.00 (0.99 – 1.01)
educ	1.67 *** (1.23 – 2.26)
reg	1.06 (0.80 – 1.39)
eco_insec	1.22 * (0.98 – 1.50)
Observations	1540
R^2 Tjur	0.065
AIC	1332.312

* *p<0.1* ** *p<0.05* *** *p<0.01*

Logit regression model on vaccine hesitancy

	low_comp
	Odds Ratios
(Intercept)	0.03 *** (0.02 – 0.07)
low_comp	1.57 *** (1.45 – 1.69)
sex	1.98 *** (1.47 – 2.69)
age	0.99 (0.98 – 1.00)
educ	1.81 *** (1.32 – 2.50)
reg	0.97 (0.72 – 1.30)
eco_insec	1.29 ** (1.03 – 1.61)
Observations	1540
R^2 Tjur	0.152
AIC	1234.701

* *p<0.1* ** *p<0.05* *** *p<0.01*

Logit regression model on vaccine hesitancy

	media
	Odds Ratios
(Intercept)	0.06 *** (0.03 – 0.12)
media	0.99 (0.75 – 1.30)
sex	1.41 ** (1.07 – 1.84)
age	1.00 (0.99 – 1.01)
educ	1.76 *** (1.31 – 2.38)
reg	1.07 (0.82 – 1.41)
eco_insec	1.26 ** (1.02 – 1.55)
Observations	1540
R^2 Tjur	0.020
AIC	1394.798

* *p<0.1* ** *p<0.05* *** *p<0.01*

Table 4: Models for nat, pray, PTV_5SM, PTV_BOI, PTV_L, risk

Logit regression model on vaccine hesitancy

	nat
	Odds Ratios
(Intercept)	0.02 *** (0.01 – 0.04)
nat	1.32 *** (1.26 – 1.39)
sex	1.17 (0.88 – 1.55)
age	1.01 ** (1.00 – 1.02)
educ	1.48 ** (1.09 – 2.03)
reg	1.00 (0.75 – 1.33)
eco_insec	1.18 (0.94 – 1.47)
Observations	1540
R^2 Tjur	0.119
AIC	1251.696

* *p<0.1* ** *p<0.05* *** *p<0.01*

Logit regression model on vaccine hesitancy

	pray
	Odds Ratios
(Intercept)	0.06 *** (0.03 – 0.13)
pray	0.79 * (0.60 – 1.04)
sex	1.37 ** (1.05 – 1.80)
age	1.00 (1.00 – 1.01)
educ	1.75 *** (1.30 – 2.37)
reg	1.06 (0.81 – 1.40)
eco_insec	1.25 ** (1.01 – 1.54)
Observations	1540
R^2 Tjur	0.022
AIC	1391.908

* *p<0.1* ** *p<0.05* *** *p<0.01*

Logit regression model on vaccine hesitancy

	PTV_5SM
	Odds Ratios
(Intercept)	0.06 *** (0.03 – 0.12)
PTV_5SM	0.98 (0.94 – 1.02)
sex	1.41 ** (1.08 – 1.84)
age	1.00 (0.99 – 1.01)
educ	1.79 *** (1.33 – 2.42)
reg	1.07 (0.82 – 1.41)
eco_insec	1.26 ** (1.02 – 1.55)
Observations	1540
R^2 Tjur	0.021
AIC	1393.970

* *p<0.1* ** *p<0.05* *** *p<0.01*

Logit regression model on vaccine hesitancy

	PTV_BOI
	Odds Ratios
(Intercept)	0.03 *** (0.02 – 0.07)
PTV_BOI	1.15 *** (1.11 – 1.19)
sex	1.43 ** (1.09 – 1.89)
age	1.01 (1.00 – 1.02)
educ	1.59 *** (1.18 – 2.16)
reg	1.03 (0.78 – 1.36)
eco_insec	1.29 ** (1.05 – 1.60)
Observations	1540
R^2 Tjur	0.054
AIC	1340.696

* *p<0.1* ** *p<0.05* *** *p<0.01*

Logit regression model on vaccine hesitancy

	PTV_L
	Odds Ratios
(Intercept)	0.04 *** (0.02 – 0.08)
PTV_L	1.13 *** (1.09 – 1.17)
sex	1.37 ** (1.04 – 1.80)
age	1.01 (1.00 – 1.02)
educ	1.60 *** (1.19 – 2.17)
reg	1.02 (0.77 – 1.35)
eco_insec	1.29 ** (1.05 – 1.60)
Observations	1540
R^2 Tjur	0.047
AIC	1350.314

* *p<0.1* ** *p<0.05* *** *p<0.01*

Logit regression model on vaccine hesitancy

	risk
	Odds Ratios
(Intercept)	0.02 *** (0.01 – 0.06)
risk	1.31 *** (1.10 – 1.57)
sex	1.40 ** (1.07 – 1.84)
age	1.00 (0.99 – 1.01)
educ	1.76 *** (1.31 – 2.38)
reg	1.05 (0.80 – 1.38)
eco_insec	1.28 ** (1.04 – 1.58)
Observations	1540
R^2 Tjur	0.027
AIC	1385.843

* *p<0.1* ** *p<0.05* *** *p<0.01*

Table 5: Models for vac_bad, vac_ob, worry, educ

Logit regression model on vaccine hesitancy

	vac_bad
	Odds Ratios
(Intercept)	0.00 *** (0.00 – 0.01)
vac_bad	3.39 *** (2.95 – 3.93)
sex	1.04 (0.75 – 1.44)
age	1.01 ** (1.00 – 1.02)
educ	1.28 (0.90 – 1.83)
reg	1.12 (0.80 – 1.55)
eco_insec	1.06 (0.82 – 1.35)
Observations	1540
R^2 Tjur	0.319
AIC	986.178

* p<0.1 ** p<0.05 *** p<0.01

Logit regression model on vaccine hesitancy

	vac_ob
	Odds Ratios
(Intercept)	0.00 *** (0.00 – 0.00)
vac_ob	5.67 *** (4.71 – 6.94)
sex	0.98 (0.67 – 1.43)
age	1.01 * (1.00 – 1.03)
educ	1.61 ** (1.07 – 2.43)
reg	1.09 (0.74 – 1.60)
eco_insec	0.79 (0.58 – 1.07)
Observations	1540
R^2 Tjur	0.509
AIC	736.441

* p<0.1 ** p<0.05 *** p<0.01

Logit regression model on vaccine hesitancy

	worry
	Odds Ratios
(Intercept)	0.05 *** (0.02 – 0.11)
worry	1.12 (0.84 – 1.50)
sex	1.41 ** (1.08 – 1.85)
age	1.00 (0.99 – 1.01)
educ	1.77 *** (1.32 – 2.39)
reg	1.07 (0.82 – 1.40)
eco_insec	1.26 ** (1.02 – 1.56)
Observations	1540
R^2 Tjur	0.021
AIC	1394.191

* p<0.1 ** p<0.05 *** p<0.01

Logit regression model on vaccine hesitancy

	educ
	Odds Ratios
(Intercept)	0.06 *** (0.03 – 0.12)
educ	1.76 *** (1.31 – 2.38)
sex	1.41 ** (1.07 – 1.84)
age	1.00 (0.99 – 1.01)
reg	1.07 (0.82 – 1.41)
eco_insec	1.26 ** (1.02 – 1.55)
Observations	1540
R^2 Tjur	0.020
AIC	1392.803

* p<0.1 ** p<0.05 *** p<0.01

8 Additional materials for scatterplots

After estimating these regression models, we exponentiated the original coefficients to work with odds ratios. Table 6 summarizes the odds ratio coefficient of each variable and reports its significance. These data, together with those reported in Table 2 of Supplement S1 (Strength and Degree Centrality), are the input for the scatterplot shown in the main article (Figure 4: Scatterplot between Strength and regression coefficient).

Table 6: summary of odds ratio coefficients

Variable	*Odds ratio*	*Significance*
vac_bad	3.3883357	Significant
vac_free	5.6708848	Significant
low_worry	1.1215813	Not significant
low_risk	1.3108932	Significant
conspiracy	5.1779681	Significant
nat	1.3207578	Significant
int_locus	0.7773671	Significant
low_col_resp	1.2379600	Significant
PTV_L	1.1313915	Significant
PTV_5SM	0.9810478	Not significant
PTV_BOI	1.1476554	Significant
distrust_sci	3.3740932	Significant
pray	0.7902768	Not significant
media	0.9895658	Not significant
low_comp	1.5665293	Significant
distrust_gov	1.4257922	Significant
distrust_inst	1.3790392	Significant

Figure 10: Scatterplot Degree and Coefficient

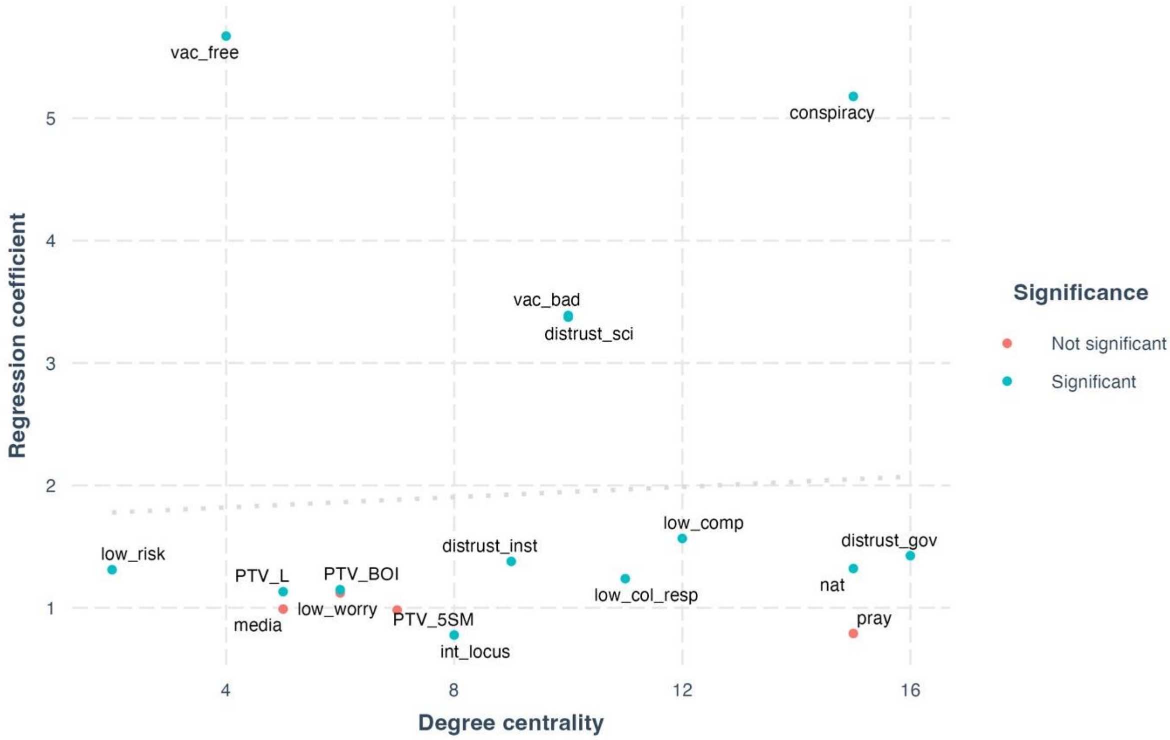


Figure 10 consists of a scatterplot between odds ratio coefficients and the degree centrality of each node. Once again, we observe a feeble relationship between degree and coefficients that vanishes when considering the role of the outliers (*vac_free*, *conspiracy*).

9 Robustness analysis

This section reports the results of several robustness analyses done with *Bootnet* (Epskamp et al., 2018), and the interested reader is suggested to consult this contribution for an in-depth discussion of the methods used. Bootstrap analyses are advised by the reporting standard for the psychological network analysis (Burger et al., 2022).

Nonparametric bootstraps allow building bootstrapped confidence intervals around each **edge** of the network. Note that if a bootstrapped CI *does* include zero, this *does not* imply that a certain edge is statistically indistinguishable from zero, as LASSO regularization already provided information about its validity (Epskamp et al., 2018). Figure 5 below reports bootstrapped confidence intervals for each edge present in the model. The red line indicates the point estimates obtained in each subsample, and the gray area is the corresponding bootstrapped CI. Horizontal lines represent network edges, ordered from the edge with the highest edge weight to the edge with the lowest edge weight. A higher resolution image is made available in the Github folder "Output/Supplement/robustness".

Confidence intervals can not be built around centrality metrics, since these values are calculated, and not estimated from the data. However, Strength centrality is affected by the uncertainty characterizing the estimation of network edges. Therefore, it is important to verify **Centrality stability** (Epskamp et al., 2018)**.** In *Bootnet*, this is done by calculating the CS coefficient. The procedure is a case-dropping bootstrap where rows are dropped randomly from the dataset. At each bootstrap iteration, the network is re-estimated, and resulting Strength scores are recorded in the process. The CS-coefficient represents the maximum proportion of cases that can be dropped, such that with 95 % probability the correlation between original centrality indices and centrality of networks based on subsets is 0.7 or higher. CS values greater than 0.5 are indicative of stable results. We obtain a very high CS score of 0.75. We show the results of the aggregated Strength stability scores in Figure 6, and we unpack these by node in Figure 7. Again, higher-resolution images are provided in the Github folder "Output/Supplement/robustness".

Finally, we execute **bootstrapped difference tests** for edges and Strength scores. This allows us to evaluate whether a certain edge is statistically stronger than another and whether a node is more important than others. Results of such a test were already reported in the main text, but Figure 8 and Figure 9 perform tests systematically for edges and centrality respectively. These plots are performed with $\alpha = 0.05$, and gray boxes indicate nodes or edges that do not differ significantly from one another. Black boxes represent nodes or edges that do differ significantly from one another instead. Blue-colored boxes in Figure 8 correspond to the intensity of the edge, and white boxes in Figure 9 show Strength scores. Note that these tests do not account for multiple testing (Epskamp et al., 2018).

Figure 5: Bootstrapped CIs

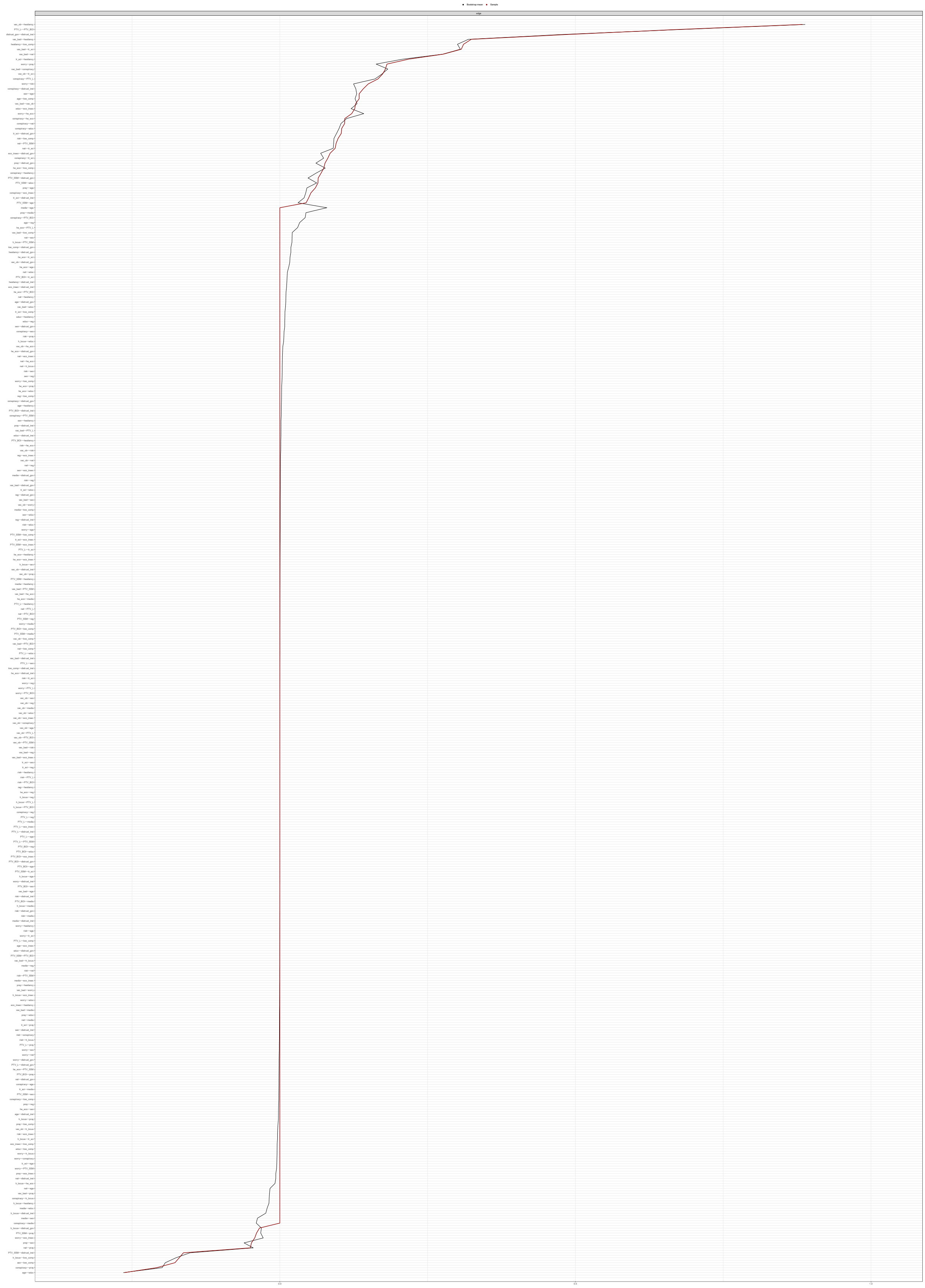


Figure 6: Aggregated Strength centrality stability

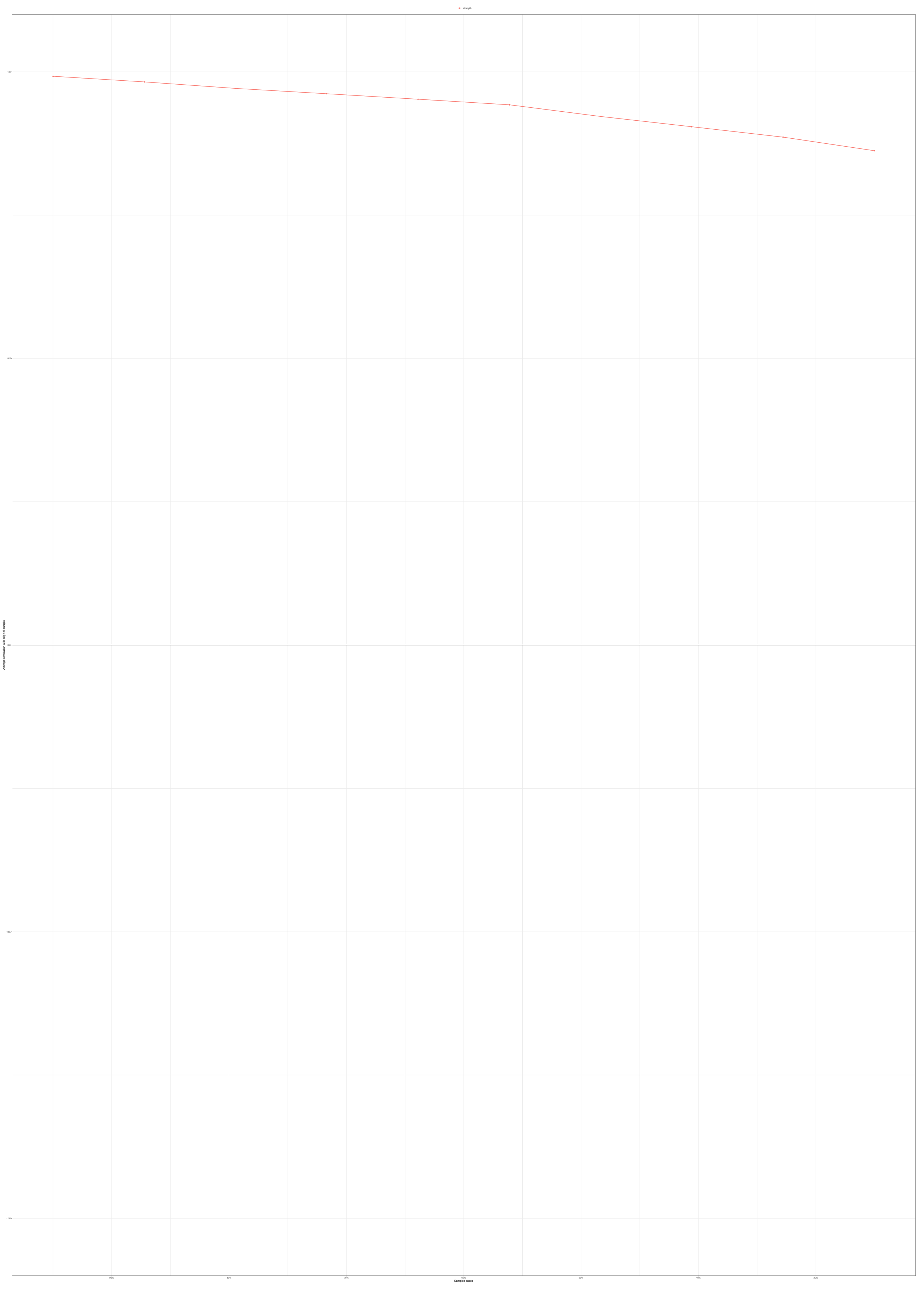


Figure 7: Strength centrality stability of each node

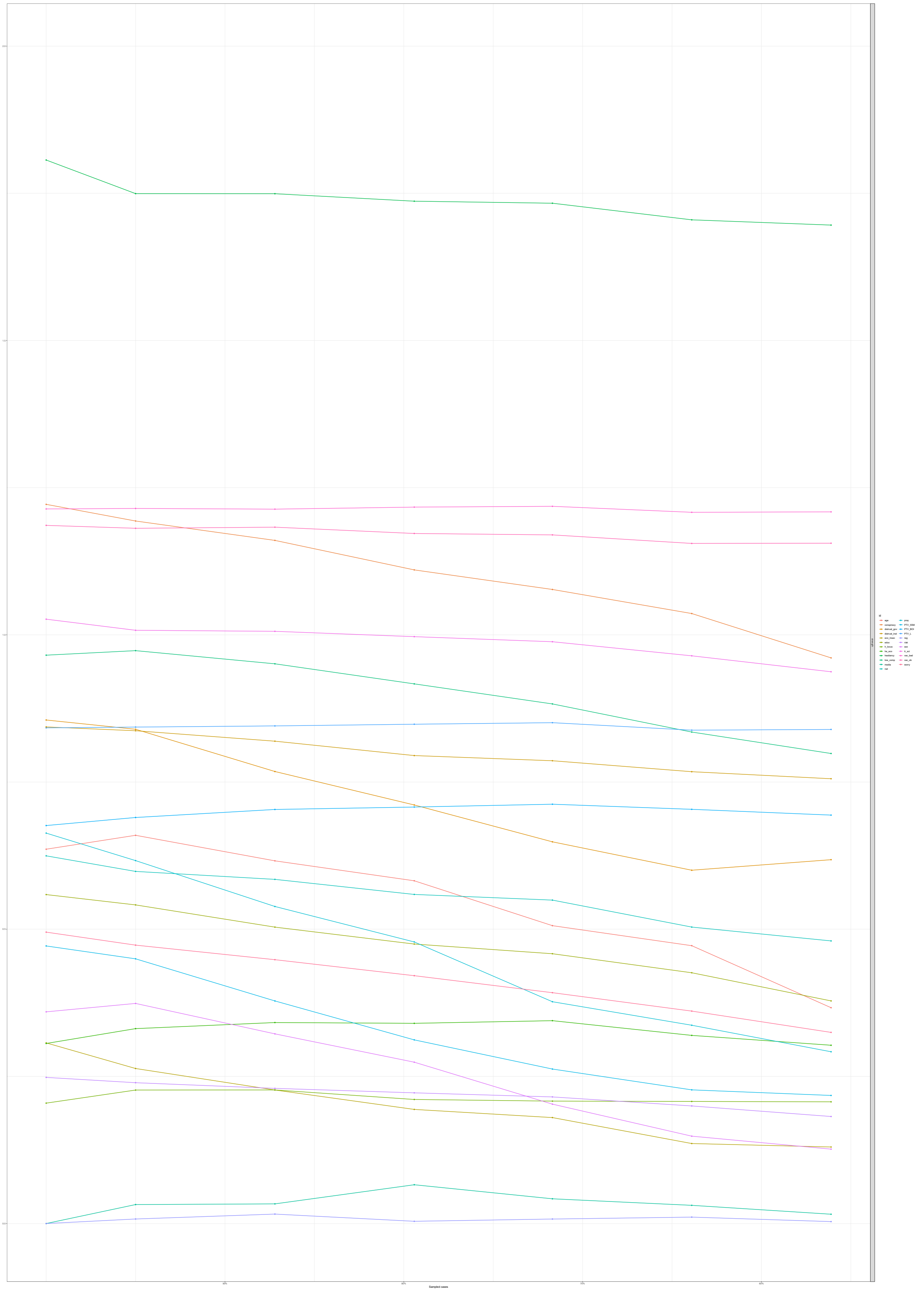


Figure 8: Edge weights bootstrapped difference tests

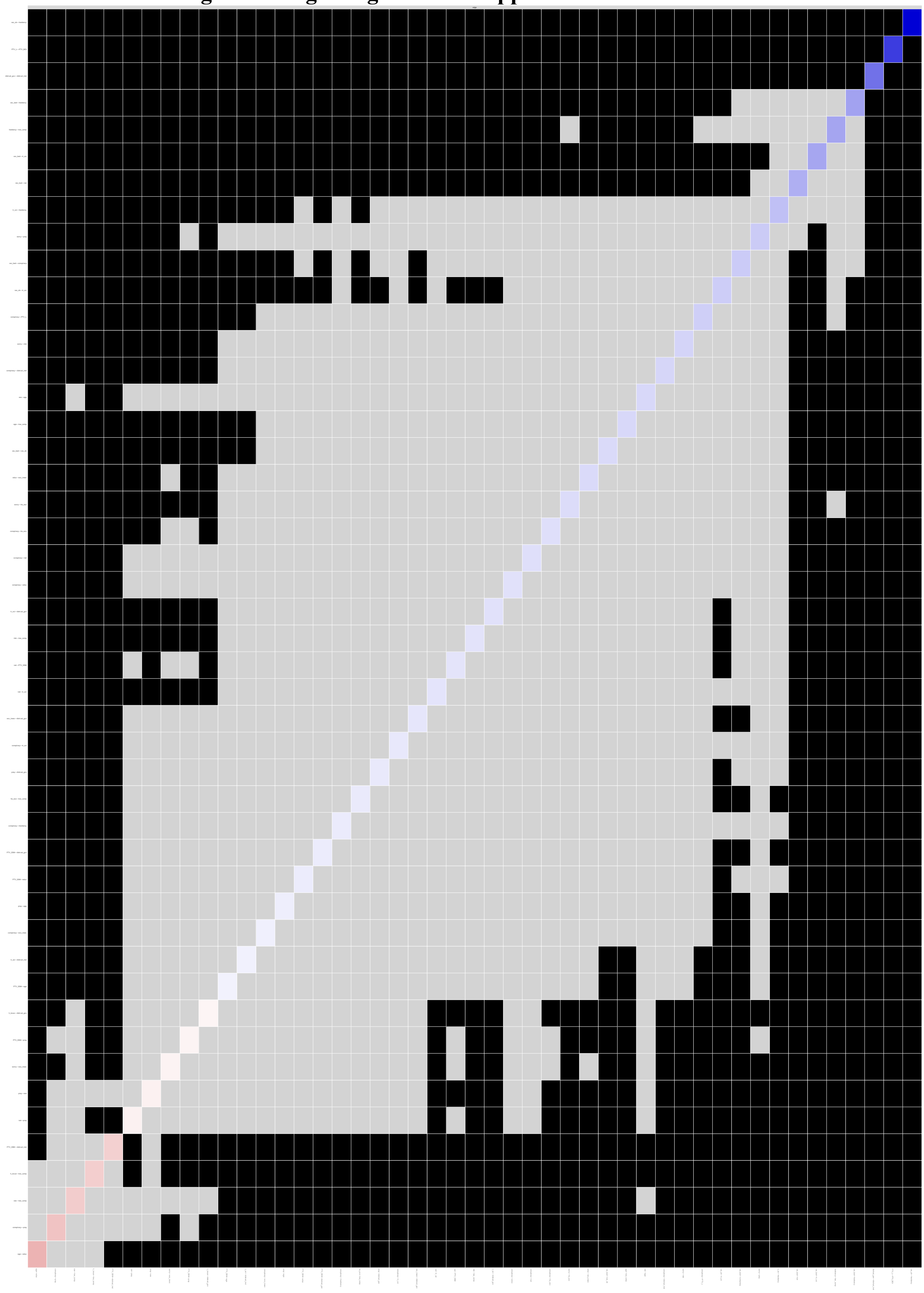
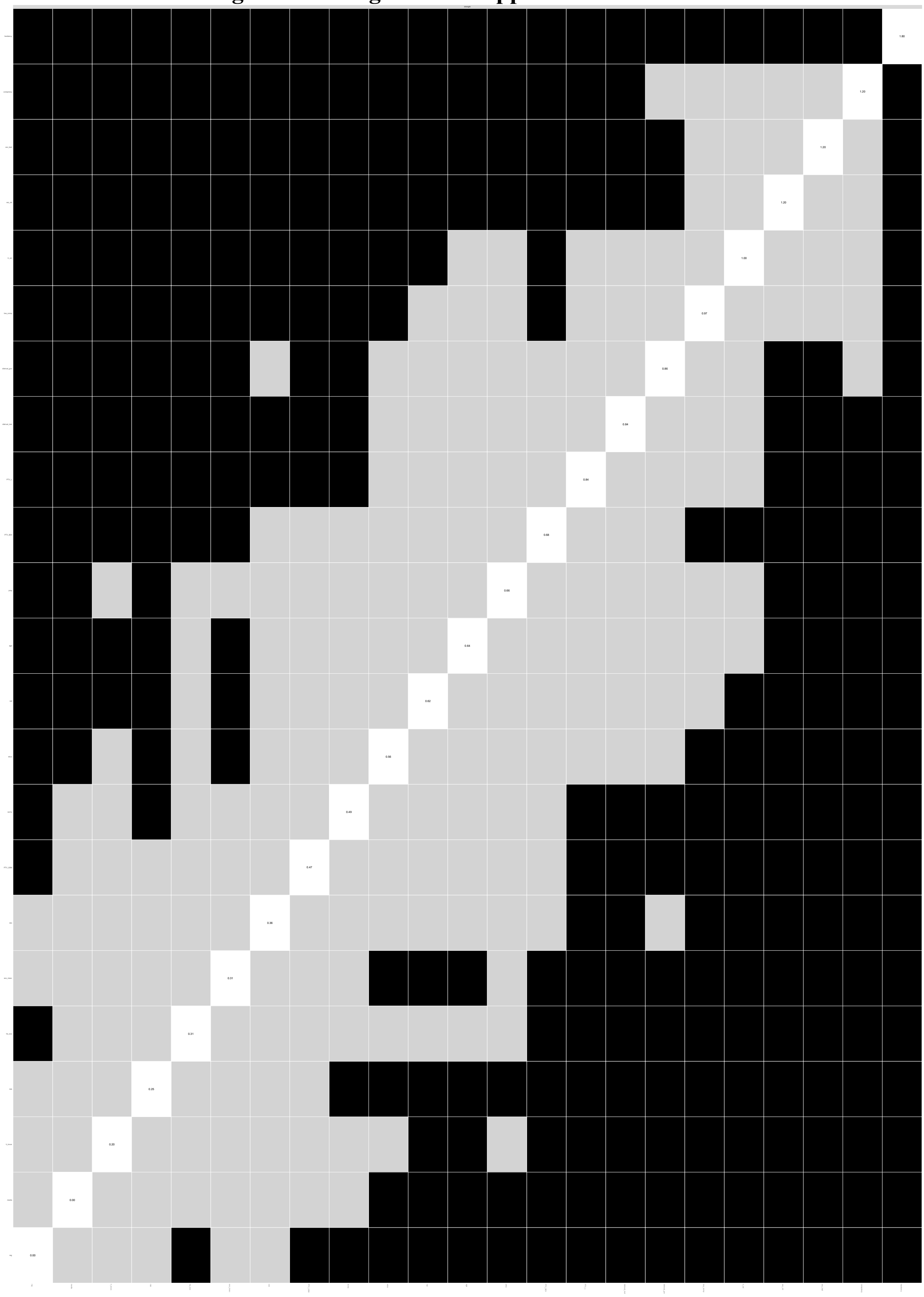


Figure 9: Strength bootstrapped difference tests

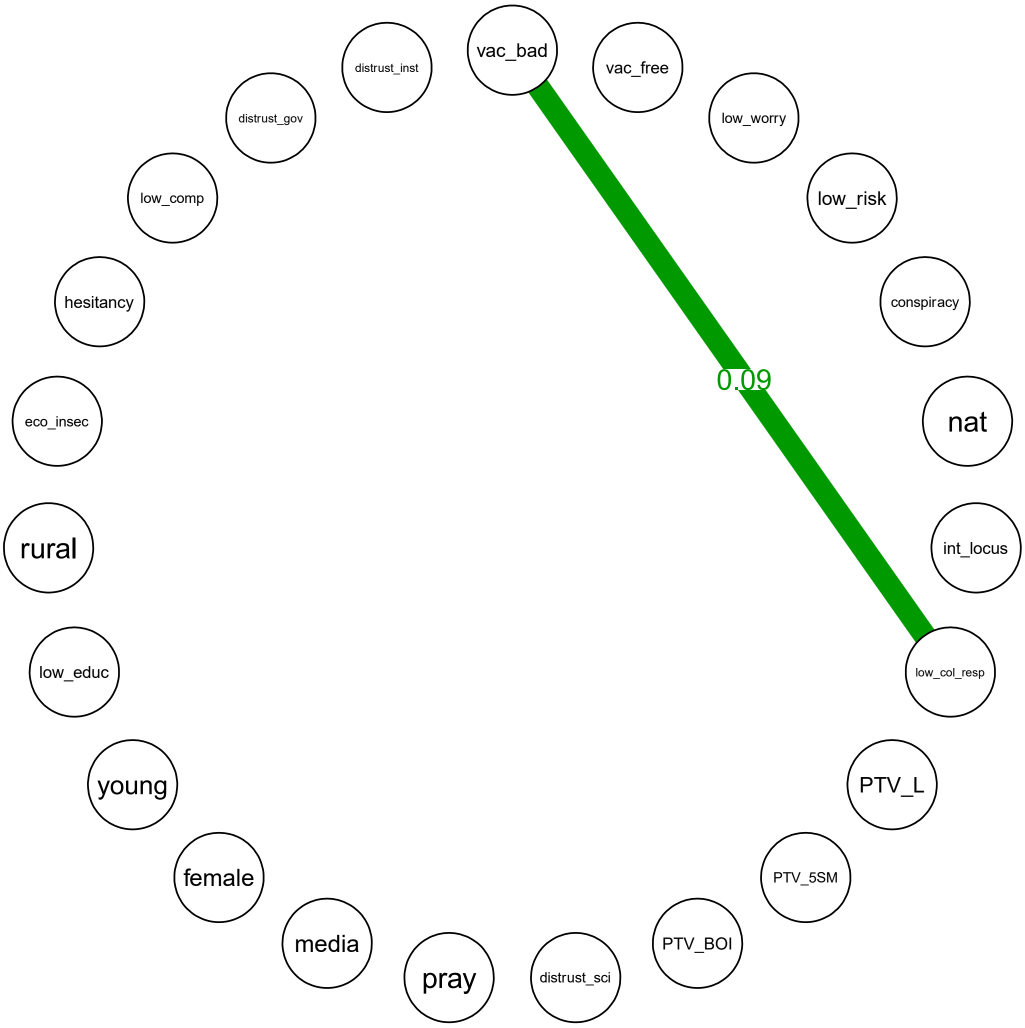


10 Replication with subsets of the sample

We have tested the stability Strength scores, and we calculated confidence intervals for each edge in the model. In this final section, we report on an additional robustness check. We randomly split the original sample (N=1540) into two partitions (N=770; N=770). We perform network estimation and centrality calculation within each subsample. We obtain two networks that we compare through the permutation-based Network Comparison Test (Borkulo et al., 2022). We show that the results of this paper are robust to this strategy since the two Broad Attitude Networks only slightly differ. We perform Network Comparison Test with Bonferroni correction, thus accounting for multiple testing.

Figure 11 on the next page offers a visualization of the results of the network comparison tests. The plot is composed of the same nodes of the Broad Attitude Network. Edges are drawn when the test finds a difference between the weighted adjacency matrixes of the two subsamples that are statistically significant. Red [Green] edges represent ties that are stronger in the first [second] subsample. The entity of these variations is reported next to each edge. These differences are very tiny since NCT only finds one edge differing in a statistically significant way between the networks estimated on the two random subsamples. Moreover, note that *hesitancy* and *conspiracy*, the nodes on which the article focuses the most, do not interact differently with other variables in the two subsamples. This reaffirms the stability of the results. In Section 2 of Supplement S2 (paragraph "Additional robustness check"), we show that no differences emerge regarding Strength centrality scores. Some point estimates do change across the networks, but these results are not statistically significant.

Figure 10: visualization of Network Comparison Test results



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